The Colours of the World's Place Names

Black, white, red, green, blue and yellow in 138,453 names from GeoNames, in 20 languages

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Abstract

Places all over the world are named after their colour: Black Creek, Río Blanco, Cerro Colorado, Kara Su, Krasnaya Gorka, Baishui. I extract colour names from GeoNames, the largest open gazetteer, with a classifier written separately for each of 20 languages or language groups, and check it on review samples of up to 50 names for each colour and language (4,370 names; 99% judged correct after tuning) and of 50 names for each rule of a second, wider pass (299 of 300 correct). The main set holds 138,453 natural features and settlements in 214 countries and territories. White is the commonest colour (26.5%), then black (23.2%) and red (20.6%); without the very numerous Chinese and Japanese names black edges ahead (27.3% against 25.7%) and blue is rare everywhere (5.5% overall). The colour depends on the kind of place: weighting languages equally, rivers are the blackest kind (37.2% black, blacker than their language's average in 14 of 17 languages) and springs the whitest (42.7%), while mountains are only weakly white. Each language has a favourite: svart in the Nordic languages, siāh in Persian, kara and ak in the Turkic languages, blanco and colorado in Spanish, bai and huang in Chinese. In the 102 countries with at least 30 names, the commonest colour is white in 35, black in 35 and red in 21. Red is partly political: 21.9% of East Slavic and 29.8% of Chinese red names are revolutionary. Names of earth, soil and clay are no redder than names in general.

Keywords: toponymy; colour terms; place names; GeoNames; basic colour terms; historical geography; maps

1 Introduction

Colour is one of the oldest ways of naming a place. A peat-dark stream, a chalk spring, a hill of red soil or a green island is named for what it looks like, and these names are among the most transparent in any language. The words for colour are also among the most studied words in linguistics: since Berlin and Kay (1969) it has been argued that languages gain their basic colour terms in a broadly fixed order, black and white (or dark and light) first, then red, then green and yellow, then blue.

This study asks how colours are used in the names of places across the world. Which colours are used most, and does that follow the order of colour terms? Do kinds of place have characteristic colours, black rivers or white mountains? Which colours does each language favour, and which country's names are blackest or whitest? And how much of the red in place names is the colour of the ground? It accompanies an interactive map of every name (https://maps.tristanhotham.com/colour-names/).

2 Data

GeoNames. GeoNames (GeoNames, 2026) is an open gazetteer of about 13 million features, each with a name, an ASCII form, alternate names in other languages and scripts, coordinates, a feature class and code (stream, mountain, populated place and so on) and a country code. It is released under CC BY 4.0. It is built from national gazetteers and its density varies greatly: the United States, Mexico, China, Norway and Iran are covered in great detail, much of Africa and South Asia thinly.

The candidates. Two searches on colour words were run over the full allCountries dump and kept 252,615 rows. The first (185,388 rows) took whole words in English and the Romance languages; the start of words in German, Dutch, the Nordic languages, Polish and Russian; whole words in the Turkic, Arabic and Persian romanisations; and the colour characters in the Chinese and Japanese alternate names. A second, wider pass (67,227 rows) added compounds and languages the first had missed: English compounds in Britain, Ireland and the former colonies (Blackburn, Whitby, Greenwich, Redcar), Turkic compounds (Aksu, Karakol, Kızılırmak, Göksu), Norwegian and Danish *kvit* and *hvid*, Mongolian (*khar*, *tsagaan*, *ulaan*, *nogoon*, *khökh*, *shar*), Hindi, Urdu and Nepali (*kala*, *safed*, *lal*, *hara*, *nila*, *pila*) and Korean. These are candidates, most of which are not colour names in the sense meant here.

Other sources. Country outlines are from Natural Earth (Natural Earth, 2026); the interactive map uses the Inked World map style (Hotham, 2026) on OpenStreetMap data served by OpenFreeMap.

3 Methods

3.1 A classifier for each language

A colour word in a place name is often something else: a surname (White's Creek, Colonia Rojas, Huangjiacun), a saint (Santa Blanca), a person (Red Nichols Lake), a state (Colorado Springs), a plant or animal (Red Oak, Blue Jay), a revolutionary slogan (Krasnyy Oktyabr'), or a different word that looks the same (Rotterdam, Belgorod's neighbour Belovo, Finnish *vita-*). The classifier therefore has a rule set for each language (Table 4). A name passes when one of its words is a colour word of a language spoken in the country where the place lies, in a form that language uses for the colour: the plain or inflected adjective, or the first part of a compound whose second part is a landscape word from a list (German *Schwarz-wald*, Norwegian *Svart-berget*, Afrikaans *Wit-rivier*, Turkish *Kızıl-ırmak*, Russian *Krasno-yarsk*). Each language has a stop-list: names after a saint, a possessive (*X's*), family-name endings (Polish *-ów*, *-owo*; Russian *-ov*, *-ovo*); in the United States, Canada and Australia, Spanish *negro* and *colorado* count only with a Spanish word beside them, since Negro Creek is named for people and Colorado City for the state. Chinese and Japanese names are read only in their own characters, from the alternate names, since romanised syllables such as *bai* or *hong* have many other meanings; Chinese names whose colour character is followed by a word for family or house (*jia*, *zhuang*) are treated as family names, and pasture names written in sound for Tibetan and Mongolian words are dropped. The rules of the second pass are stricter still (Table 1). An English compound must end in a landscape element of English place names (*-burn*, *-pool*, *-by*, *-church*, *-wich*, *-water*); *Red-* takes a short list (*-hill*, *-car*, *-cliff*), since it is often Old English *hrēod*, reed, or Cornish *rhyd*, ford (Redruth, Redbourn, Redmire), and bare *Whit-* counts only in Britain and Ireland (Whitby, Whitchurch), since Whitney and Whittington come from the personal name Hwita; in North America, Australia, New Zealand and South Africa common surnames (Blackburn, Greenwood, Whitley) and towns named after people (Greenville) are

Table 1: The second pass: candidate rows found by each search, names accepted into the main set, and review sample judged correct.

Rule	Candidates	Accepted	Correct
Turkic compounds	32,503	9,395	50 of 50
English compounds	24,626	2,254	49 of 50
Norwegian kvit, Danish hvid	1,854	1,374	50 of 50
Mongolian	2,809	793	50 of 50
Hindi, Urdu and Nepali	2,546	238	50 of 50
Korean	2,889	184	50 of 50

stopped. A Turkic stem must start the word and be followed by a landscape word (*Ak-su*, *Kara-kol*, *Kızıl-ırmak*), with stops for *karaca* (roe deer), *akar* (flowing) and the like. Mongolian words count as whole words or hyphenated compounds, never Russian *shar*, a strait. Hindi and Urdu words count only beside a landscape word (Kala Pani, Lal Tibba), which keeps out personal names (Lal Khan, Moti Lal) and the goddess Kali but also most red and white names. Korean names count only where GeoNames gives the colour hanja (紅 for hong, as hong is also the family name 洪) or where a hangul compound is unambiguous (흑, black; 백석, white stone; 청룡, blue dragon; 황토, yellow earth). The single East Asian character 青 (qing in Chinese, ao in Japanese), which covers blue and green, is kept as its own category, blue-green.

Two flags are kept rather than dropped. English names in which the colour belongs to a plant or animal (Red Oak, White Pine, Blackbird) are flagged and left out of the main set (2,900 names). Red names from the Soviet and Chinese revolutionary vocabulary (with *Oktyabr'*, *Zvezda*, *Partizan*, 红旗 Red Flag, 红星 Red Star and the like) are flagged as political and kept, since they are red names, but reported apart.

3.2 Kinds of place and the main set

GeoNames feature codes are grouped into nine kinds of place: rivers, streams and wadis; lakes, ponds and marshes; springs and wells; seas, bays and coasts; mountains and hills; rocks, cliffs and islands; valleys, gorges and passes; plains, deserts, woods and other land; and towns and villages. Buildings, farms, mines, hotels, parks, roads, administrative areas and undersea features (35,478 accepted names) are left out of the main set, because they often take a colour from an owner, a brand or another place.

3.3 Review

For every language and colour I read a sample of up to 50 accepted names, drawn in a fixed pseudo-random order (an MD5 hash of the GeoNames identifier), and judged whether the word is the colour there. The samples were used to tune the rules, and the final precision is measured on the samples drawn after tuning, so it is optimistic: names the tuning removed were found in these samples. In all I read 4,459 different names.

3.4 Comparisons

Shares of each colour are given for all names pooled, and, where noted, with each language weighted equally (languages with at least 30 names of the kind of place in question), so that the 29,530 Chinese names, most of them villages, do not dominate. Countries are compared on the most common colour among their names and on the ratio of black to white names, with half

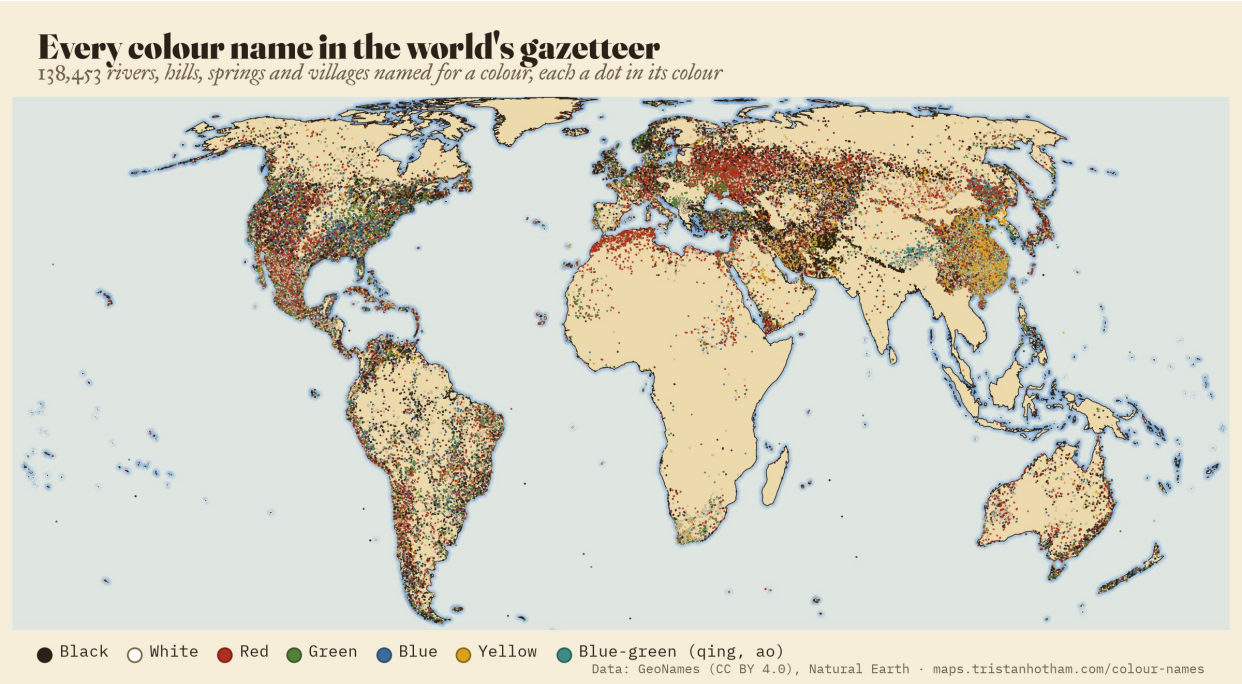


Figure 1: Every colour name in the main set, a dot in its colour (Equal Earth projection).

a name added to each count. Names of earth, soil, sand and clay are found by words in the name (*tierra*, *terra*, *barro*, *glina*, *khāk*, *kum* and others, and the characters 土 and 泥).

4 Results

4.1 The whole map

The main set holds 138,453 names in 214 countries and territories (Figure 1): 77,115 natural features and 61,338 settlements. The density of dots follows the density of the gazetteer: the United States, Mexico, China, Norway, Russia, Kazakhstan, Turkey and Iran stand out. The second pass contributes 14,238 names, 9,395 of them Turkic compounds and 2,254 English ones.

4.2 Which colours

White is the commonest colour (26.5%), then black (23.2%), red (20.6%), yellow (10.7%), green (10.5%), blue-green (3.0%) and blue (5.5%). Yellow's place owes everything to China: without Chinese and Japanese names the order is black 27.3%, white 25.7%, red 21.3%, green 13.1%, blue 6.9% and yellow 5.6%. Black, white and red together make up more than half of the colour names of every language with at least 300 names, matching the first stages of Berlin and Kay (1969); blue is never among the three commonest of the six colours in any such language (Figure 2).

4.3 Which places

With languages weighted equally (Figure 3, Table 2), rivers and streams are the blackest kind of place (37.2% black against 29.1% white), and rivers are blacker than their language's average in 14 of 17 languages. Springs and wells are the whitest (42.7%). Red reaches its highest share on

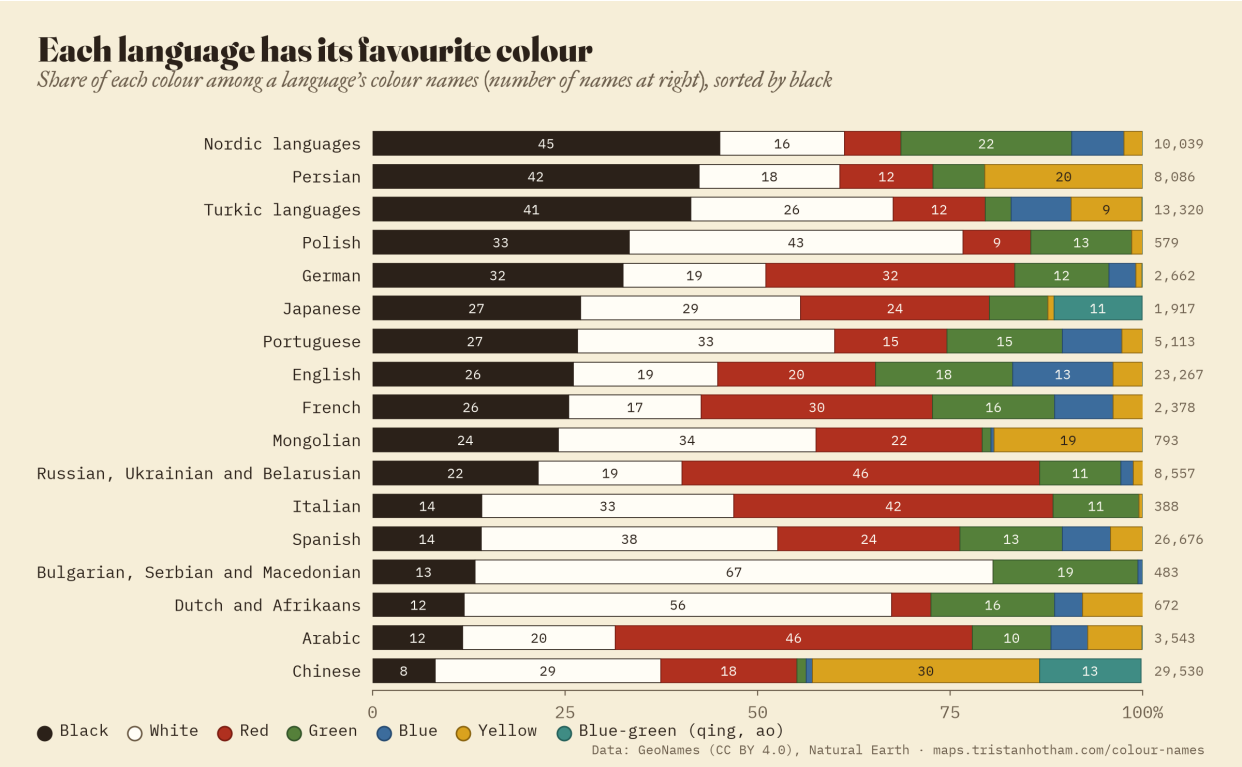


Figure 2: Share of each colour among the colour names of each language with at least 300 names.

Table 2: Colour by kind of place: names in the main set, and the share of each colour (%) with languages weighted equally. BG: blue-green.

Kind	Names	Black	White	Red	Green	Blue	Yellow	BG
Rivers, streams and wadis	19,269	37.2	29.1	15.0	6.6	3.8	7.4	0.9
Lakes, ponds and marshes	10,848	30.1	29.2	16.8	9.9	6.4	6.5	1.0
Springs and wells	3,384	14.8	42.7	21.5	6.6	7.1	7.2	0.0
Seas, bays and coasts	1,793	28.1	24.8	16.8	11.5	4.5	6.0	8.2
Mountains and hills	25,784	27.6	28.4	23.4	9.1	4.0	5.9	1.6
Rocks, cliffs and islands	6,265	26.0	24.7	24.2	11.7	2.9	7.5	3.0
Valleys, gorges and passes	4,971	27.9	27.6	20.6	10.8	4.2	8.9	0.0
Plains, deserts, woods and areas	4,801	22.7	29.8	22.1	10.8	3.4	10.0	1.2
Towns and villages	61,338	23.4	28.0	20.3	14.9	3.6	6.7	3.1

rocks, cliffs and islands. Mountains are white more often than black (28.4% against 27.6%, with red at 23.4%), but they are whiter than their language's average in only 7 of 16 languages; they are whiter than average in Chinese, Japanese, Mongolian, Dutch and Afrikaans, Russian, the Nordic languages and French, but not in Spanish, German, English, Persian or the Turkic languages. The ratio of black to white names is 1.28 for rivers, 1.03 for lakes and 1.13 for seas and coasts, but 0.35 for springs and 0.84 for settlements.

4.4 Languages and their favourite colours

Table 3 sets out the colour mix of each language. In the Nordic languages 45.1% of colour names are svart, black, mostly lakes, tarns and fells; in Persian and Dari 42.4% are siāh, and Kūh-e Sīāh, Black Mountain, occurs 284 times. In the Turkic languages kara and ak make up 67.7% of colour names, and Karasu, black water, occurs 239 times. Mongolian favours white (33.5%, *tsagaan*), then black (24.1%, *khar*, mostly in Buryatia) and red (21.6%, *ulaan*). Arabic (46.4%) and the East

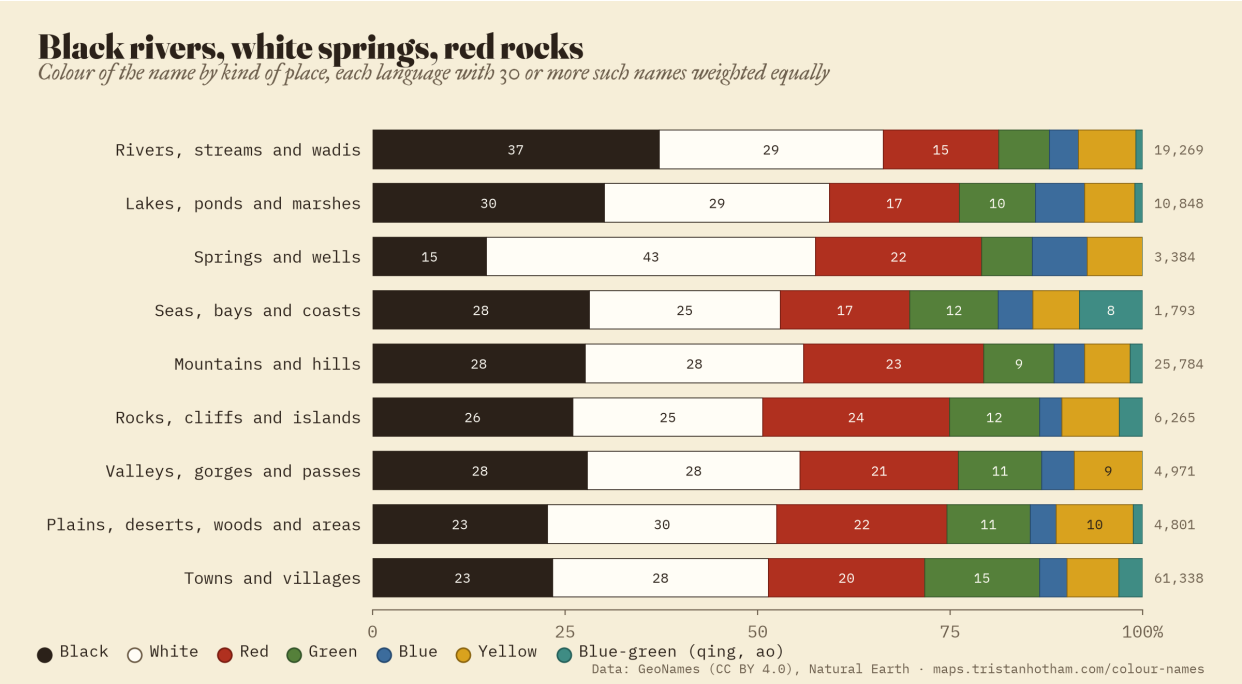


Figure 3: Colour of the name by kind of place, languages with at least 30 names of that kind weighted equally.

Table 3: Colour names by language: names in the main set, share of each colour (%) and the share of review-sample names judged correct (%).

Language	Names	Black	White	Red	Green	Blue	Yellow	BG	Correct
Chinese	29,530	8	29	18	1	1	30	13	97
Spanish	26,676	14	38	24	13	6	4	0	99
English	23,267	26	19	20	18	13	4	0	97
Turkic languages	13,320	41	26	12	3	8	9	0	99
Nordic languages	10,039	45	16	7	22	7	2	0	99
Russian, Ukrainian and Belarusian	8,557	22	19	46	11	2	1	0	100
Persian	8,086	42	18	12	7	0	20	0	100
Portuguese	5,113	27	33	15	15	8	3	0	99
Arabic	3,543	12	20	46	10	5	7	0	99
German	2,662	32	19	32	12	4	1	0	100
French	2,378	26	17	30	16	8	4	0	99
Japanese	1,917	27	29	24	8	0	1	11	99
Mongolian	793	24	34	22	1	0	19	0	nan
Dutch and Afrikaans	672	12	56	5	16	4	8	0	100
Polish	579	33	43	9	13	0	1	0	100
Bulgarian, Serbian and Macedonian	483	13	67	0	19	1	0	0	100
Italian	388	14	33	42	11	0	0	0	98
Hindi, Urdu and Nepali	238	83	2	8	4	1	1	0	nan
Korean	184	15	15	0	0	0	38	32	nan
Romanian	28	0	0	0	100	0	0	0	100

Slavic languages (46.4%) are the reddest: aḥmar and ḥamrāʾ for desert rock and sand, krasnyy for villages. Spanish is the whitest of the large languages (38.5% white), and the commonest colour name in the world is the Spanish Cerro Colorado (1,084 names), followed by Cerro Blanco (748) and Cerro Negro (531). Unlike most languages, Spanish and Portuguese rivers are whiter than black (36.4% against 21.9%; 428 Río Blanco against 309 Río Negro). In Chinese, bai (white, 白) and huang (yellow, 黄) make up 58.8% of colour names (Figure 4).

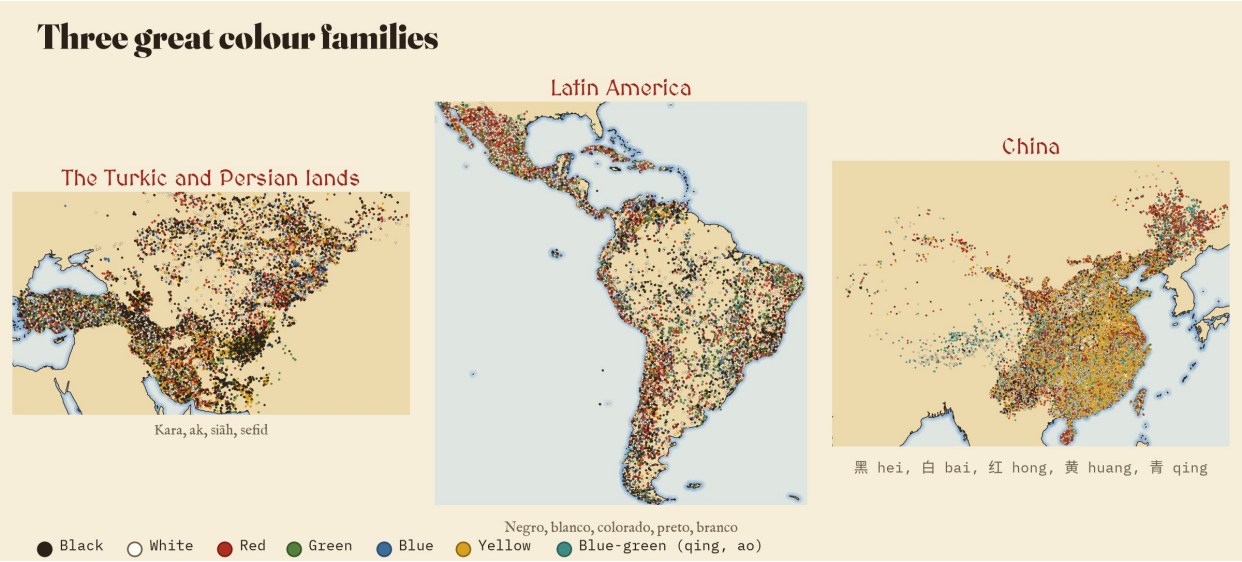


Figure 4: Three colour families: the Turkic and Persian lands, Latin America and China (plate carrée at each panel's middle latitude).

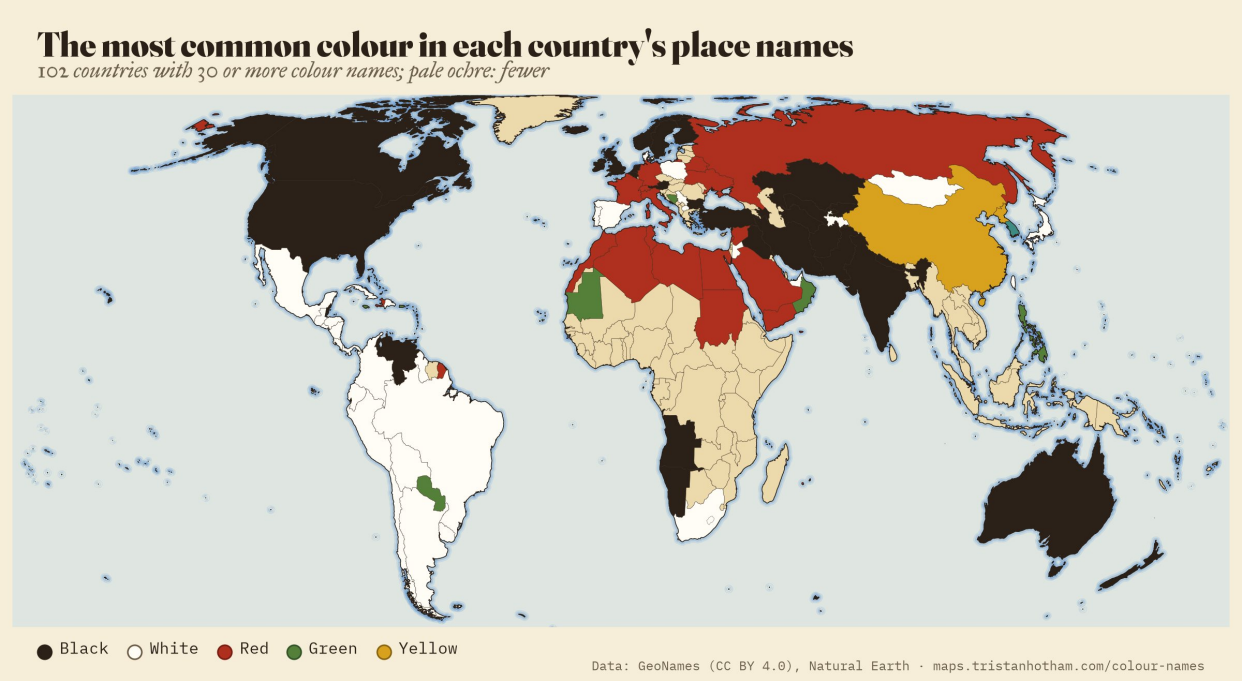


Figure 5: The commonest colour among each country's colour names, countries with at least 30 names.

4.5 Countries

In the 102 countries with at least 30 names, the commonest colour is white in 35, black in 35, red in 21, green in 8, yellow in 2 and blue-green in 1 (South Korea); blue leads in none (Figure 5). Black leads in the English-speaking countries, the Nordic countries, Turkey, Central Asia, Iran and Afghanistan; white in most of Latin America, Mongolia, Serbia, North Macedonia, Poland and South Africa; red in Russia, Belarus and much of the Arab world; yellow in China. By the ratio of black to white names (Figure 6), 41 countries have more black than white names and 57 more white than black. With *kvit* from the second pass, Norway's ratio is 2.8 to one. The most extreme ratios now come from the narrow Hindi and Urdu rule: India (32.6) and Pakistan (30.6) look very black because nearly all the names it accepts are *kala*.

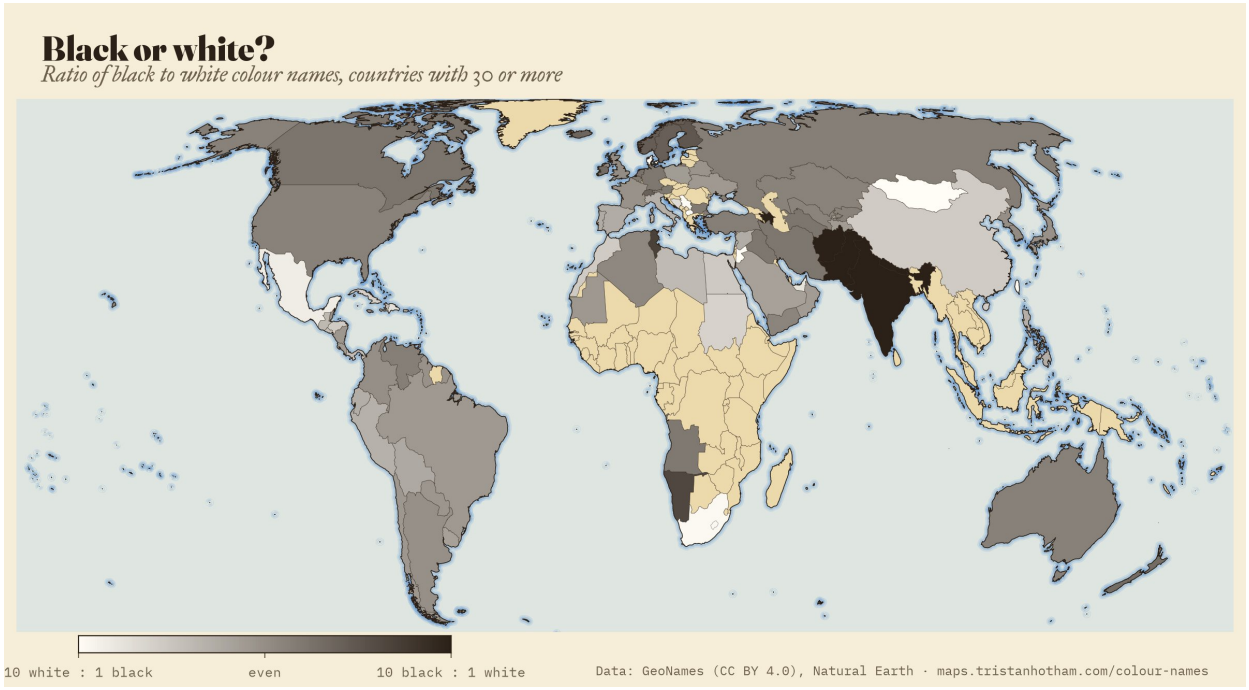


Figure 6: Ratio of black to white colour names, countries with at least 30 names (log scale).

4.6 Hotspots

On a two-degree grid the densest cell of black names is in central Norway (350 names), of white and red names in central Mexico (342 and 219), of green and blue names in western Norway, and of yellow names in southern China (288). These follow the gazetteer's density as much as naming habits.

4.7 Red: revolution and earth

Of 3,974 red names in Russian, Ukrainian and Belarusian, 871 (21.9%) are revolutionary; of 5,241 Chinese red names, 1,561 (29.8%). Without them, red would fall from 46.4% to 40.4% of East Slavic colour names. Some of the remaining East Slavic red is not colour at all: *krasnyy* long meant beautiful, so a Krasnaya Polyana may be a fair glade. Of 3,735 names of earth, soil, sand and clay, 763 (20.4%) are red, close to the share of red among all names (20.6%); 1,247 are white (Tierra Blanca alone 406 times) and 1,200 yellow (Figure 7).

4.8 Precision

After tuning, 4,327 of 4,370 sampled names were judged correct (99%; 0.98 when each language and colour cell is weighted by its number of names). The lowest cell with at least 20 names sampled reaches 0.92. English (0.97) and Chinese (0.97) are the weakest languages, for family names and names written in sound (Figure 8).

5 Limitations

First, recall is limited by the searches that produced the candidates and by the strictness of the rules. The second pass filled the largest gaps of the first (English and Turkic compounds, Norwe-

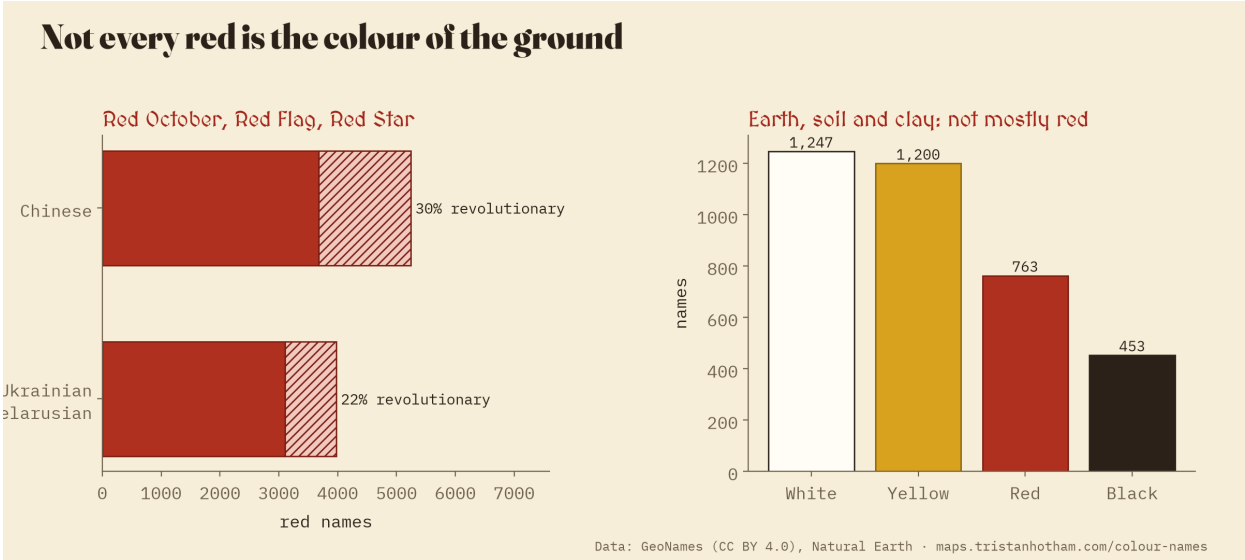


Figure 7: Left: revolutionary red among East Slavic and Chinese red names. Right: colours of names of earth, soil, sand and clay.

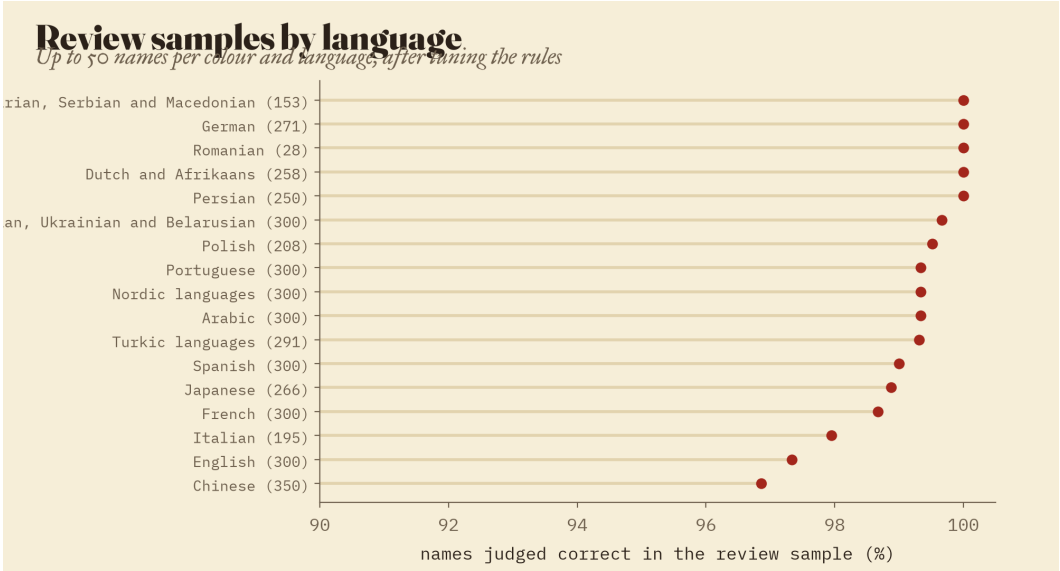


Figure 8: Names judged correct in the review samples, by language (names sampled in brackets).

gian *kvit*, Mongolian), but Korean, Hindi and Urdu are covered only where the meaning is clear, which favours some colours (black in South Asia); Afrikaans *rooi* and *swart*, Persian *kabud*, and whole languages (Hungarian, Finnish, Greek, Malay and Indonesian, Pashto and the languages of Africa) are missing. The second pass was reviewed with only 50 names per rule. Second, GeoNames is uneven: counts follow the density of national gazetteers and include duplicates of the same feature. Third, precision is measured on samples that also guided the rules, so it is optimistic, and some judgements are uncertain: whether a Spanish *Loma Los Negros* names the colour or people, or a Chinese 黄 village a family. Fourth, a colour word does not tell us why the place was so named: Black can be peat, shade, a forest, a fearful reputation or a direction (in the Turkic tradition black has been read as north and white as west, Wikipedia, 2026). Fifth, the six colours and blue-green are this study's categories, not the languages' own.

6 Conclusion

The colours of the world's place names follow the oldest colour words: white, black and red make up most of them, blue is rare. Each kind of place has a leaning, black for rivers and dark water, white for springs, red for rock and bare ground, and each language has its favourite word, from svart and siāh to blanco and huang. A good part of the red, though, is not the colour of the land but of twentieth-century flags.

Data and code availability

The interactive map is at https://maps.tristanhotham.com/colour-names/. Data and code are at https://github.com/Tristan-Hotham/colour-names and archived with this paper at https://doi.org/10.5281/zenodo.24043682. Derived data are released under CC BY 4.0 and code under the MIT licence. Place names are from GeoNames (CC BY 4.0); country outlines from Natural Earth (public domain); the map style uses OpenStreetMap data (ODbL).

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

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A The classifier's colour words

Table 4: Colour words (dictionary forms; inflected and compound forms are also matched) and the countries where each rule applies.

Language	Words	Where
English	black: black; white: white; red: red; green: green; blue: blue; yellow: yellow	anywhere
Spanish	black: negro; white: blanco; red: rojo, colorado; green: verde; blue: azul; yellow: amarillo	anywhere outside the Portuguese and Italian areas; in the US, Canada and Australia only with a Spanish word beside it
Portuguese	black: preto, negro; white: branco; red: vermelho; green: verde; blue: azul; yellow: amarelo	Brazil, Portugal, Angola, Mozambique, Cape Verde and other Lusophone countries
French	black: noir; white: blanc; red: rouge; green: vert; blue: bleu; yellow: jaune	anywhere
Italian	black: nero; white: bianco; red: rosso; green: verde; yellow: giallo	Italy, San Marino, Switzerland
Romanian	green: verde	Romania, Moldova
German	black: schwarz; white: weiss; red: rot; green: grün; blue: blau; yellow: gelb	Germany, Austria, Switzerland, Liechtenstein, Luxembourg, Italy, Belgium, Namibia
Dutch and Afrikaans	black: zwart; white: wit; red: rood; green: groen; blue: blauw; yellow: geel	Netherlands, Belgium, Suriname, the Dutch Caribbean, South Africa, Namibia
Nordic languages	black: svart; white: hvit, vit; red: rød, röd; green: grønn, grön; blue: blå; yellow: gul	Norway, Sweden, Denmark, Finland, Iceland, the Faroes, Greenland
Polish	black: czarny; white: biały; red: czerwony; green: zielony; blue: niebieski; yellow: żółty	Poland
Russian, Ukrainian and Belarusian	black: chërnyy; white: belyy; red: krasnyy; green: zelënyy; blue: siniy; yellow: zhëltyy	the former Soviet Union and Mongolia
Bulgarian, Serbian and Macedonian	black: cheren; white: bel; green: zelen; blue: sin	Bulgaria, Serbia, North Macedonia, Bosnia, Montenegro, Kosovo
Turkic languages	black: kara, qara; white: ak, aq; red: kızıl, kyzyl; green: yeşil; blue: gök, kök; yellow: sarı, sary	Turkey, the Caucasus, Central Asia, Russia, Iran, Afghanistan, China, Mongolia and the Balkans
Arabic	black: aswad, sawdā'; white: abyaḍ, bayḍā'; red: aḥmar, ḥamrā'; green: akhḍar, khaḍrā'; blue: azraq, zarqā'; yellow: aṣfar, ṣafrā'	the Arab world, Chad, Mali, Niger, Iran, Turkey (with an Arabic word beside it)
Persian	black: siāh; white: sefid; red: sorkh; green: sabz; yellow: zard	Iran, Afghanistan, Tajikistan, Pakistan
Chinese	black: 黑 hei; white: 白 bai; red: 红 hong; green: 绿 lü; blue: 蓝 lan; yellow: 黄 huang; grue: 青 qing	China, Taiwan, Hong Kong, Macau (characters)
Japanese	black: 黒 kuro; white: 白 shiro; red: 赤 aka; green: 緑 midori; yellow: 黄 ki; grue: 青 ao	Japan (characters)
Mongolian	black: khar; white: tsagaan; red: ulaan; green: nogoon; blue: khökh; yellow: shar	Mongolia, Inner Mongolia, Buryatia, Kalmykia, Tuva
Hindi, Urdu and Nepali	black: kala; white: safed; red: lal; green: hara; blue: nila; yellow: pila	India, Pakistan, Nepal, Bangladesh (beside a landscape word)
Korean	black: 黑 heuk; white: 白 baek; red: 紅 hong; green: 綠 nok; yellow: 黃 hwang; grue: 靑 cheong	South and North Korea (hanja, or unambiguous hangul compounds)

B The most repeated colour names

Table 5: The 25 most repeated colour names in the main set.

Name	Colour	Language	Places
Cerro Colorado	red	Spanish	1,084
Cerro Blanco	white	Spanish	748
Cerro Negro	black	Spanish	527
Río Blanco	white	Spanish	420
Red Hill	red	English	415
Tierra Blanca	white	Spanish	406
Agua Blanca	white	Spanish	391
Casa Blanca	white	Spanish	378
Black Creek	black	English	375
Chërnaya	black	Russian, Ukrainian and Belarusian	371
Palo Blanco	white	Spanish	359
Black Mountain	black	English	351
Cerro Verde	green	Spanish	300
Arroyo Blanco	white	Spanish	285
El Colorado	red	Spanish	284
Kūh-e Sīāh	black	Persian	284
Río Negro	black	Spanish	278
Blue Lake	blue	English	272
Black Lake	black	English	263
Green Island	green	English	252
Arroyo Colorado	red	Spanish	247
Peña Blanca	white	Spanish	242
Karasu	black	Turkic languages	239
Loma Blanca	white	Spanish	233
Ozero Beloye	white	Russian, Ukrainian and Belarusian	233